

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2016

Mathematics

Paper 1
Ordinary Level

Time: 2 hours

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2			
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1

(Suggested maximum time: 10 minutes)

- (a)** Find the value of each of the following.

(i) $\frac{24.6 + 5.8}{2^2}$

[illegible]

(ii) $87.4 - \sqrt{158.76} \times 4.5$

[illegible]

- (b) (i)** Write the number **six thousand, two hundred and thirty seven** in figures.

[illegible]

- (ii) Write the number **35 410** in words.

[illegible]

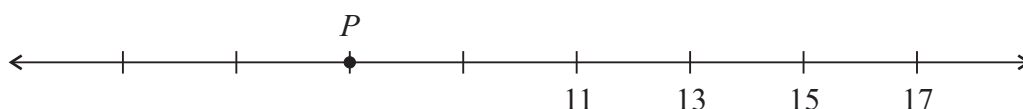
- (iii)** Write down the value of 7 in the number **6 725**.

[illegible]

- (iv) Write the number **49 449** rounded to the nearest thousand.

[illegible]

- (c) What number does the point P represent on the number line below?


$$P =$$

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Question 2

(Suggested maximum time: 10 minutes)

Patrick owns a men's clothes shop.
He buys 50 shirts costing €12.50 each.



- (a)** Calculate the total cost of the 50 shirts.

[illegible]

Patrick wishes to make a profit of 60% on the cost price of the shirts.

- (b)** Find the selling price he must charge for each shirt.

[illegible]

During a sale, Patrick reduces the price of the shirts by 55%.

- (c) Calculate the sale price of each shirt.

[illegible]

Patrick sold 20 of the shirts at the full price and the remainder of the shirts at the sale price.

- (d)** Has Patrick made a profit or a loss on the sale of all the shirts? Give a reason for your answer.

[illegible]

Question 3 (Suggested maximum time: 10 minutes)

Question 3 (Suggested maximum time: 10 minutes)

- (a)** The number 9 is half way between 4 and 14, as shown below.

14

Write down the number that is half way between each of the following.

- | | | | |
|-----|----|--|----|
| (i) | 40 | | 80 |
|-----|----|--|----|

[illegible]

- | | | | |
|------|---------|--|---------|
| (ii) | 200 000 | | 300 000 |
|------|---------|--|---------|

[illegible]

- | | | | |
|-------|-----|--|-----|
| (iii) | 8.5 | | 8.9 |
|-------|-----|--|-----|

[illegible]

- (b)** Write down the two consecutive whole numbers between which $\sqrt{40}$ lies.

$$\sqrt{40}$$
[illegible]

- (c)** Indicate, using a $>$ or $<$ symbol, which of the two fractions below is bigger.

$$\frac{3}{5}$$

Using the two grids below, or otherwise, explain your answer.

[illegible]

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Question 4

(Suggested maximum time: 10 minutes)

- (a) List the elements of the following sets.

- (i) A is the set of odd numbers between 2 and 14.

$$A = \left\{ \begin{array}{ccccc} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{array} \right\}$$

- (ii) B is the set of factors of 15.

$$B = \left\{ \begin{array}{c} \text{ } \end{array} \right\}$$

- (iii)** C is the set of prime numbers between 1 and 14.

$$C = \left\{ \begin{array}{c} \text{ } \end{array} \right\}$$

- (b) $x \in A \cap B$.

Write down all of the possible values of x .

Answer: _____, _____.

- (c) David says that $B \cap C = \emptyset$. Is he correct?

Give a reason for your answer.

[illegible]

- (d) As part of a survey, the students in a class were asked if they play football or basketball. These are the results:

15 students play football.

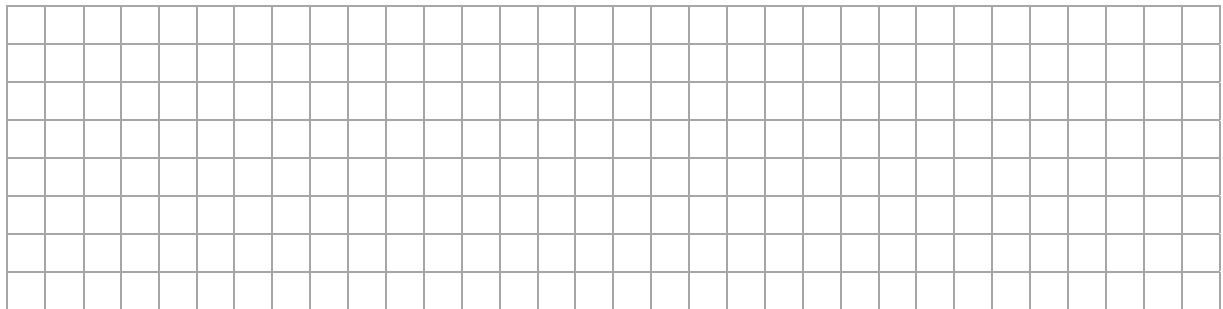
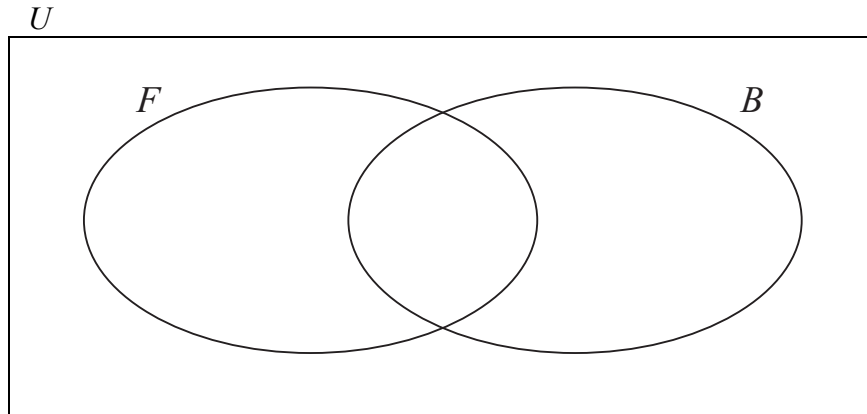
11 students play basketball.

7 students play football **and** basketball.

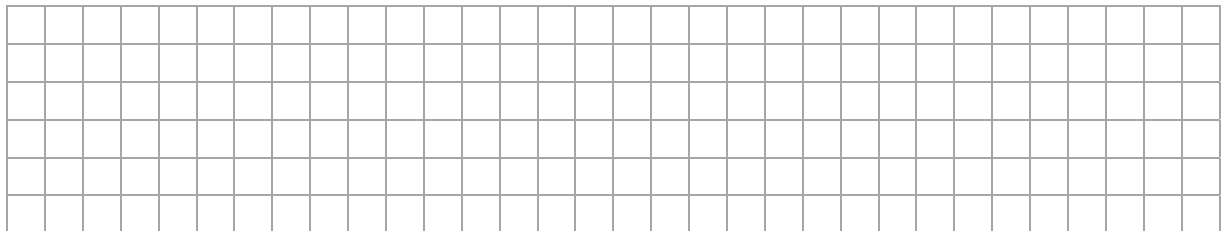
6 students do **not** play football **or** basketball.



- (i) Use these results to fill in the Venn diagram below.



- (ii) How many students are in the class?



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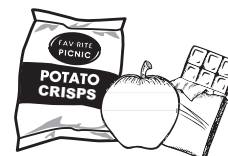
Question 5

(Suggested maximum time: 10 minutes)

Kate has organised treat bags for children calling to her house at Halloween.

Each treat bag contains:

- 1 packet of crisps,
- 1 bar of chocolate,
- 1 apple.



Kate buys some multi-packs of crisps. Each **multi-pack** contains **6 packets of crisps** and costs €2.70.

She buys some boxes of chocolate bars. Each **box** contains **18 chocolate bars** and costs **€4.50**.

She also buys some trays of apples. Each **tray** contains **12 apples** and costs **€3.60**.

- (a)** Calculate the total cost of **one** treat bag.

Contents of Treat Bag	Cost
1 packet of crisps	
1 bar of chocolate	
1 apple	
Total Cost	€

[illegible]

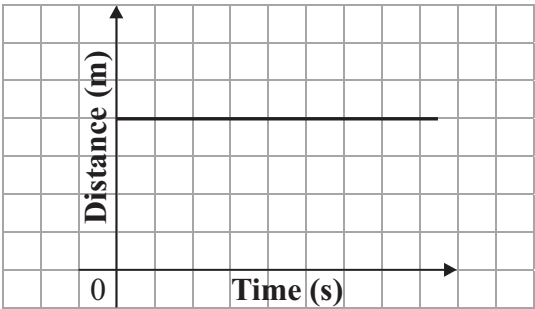
- (b)** When Kate makes up the treat bags, there are no crisps, chocolate bars or apples left over. Find the **minimum number** of treat bags that Kate makes.

[illegible]

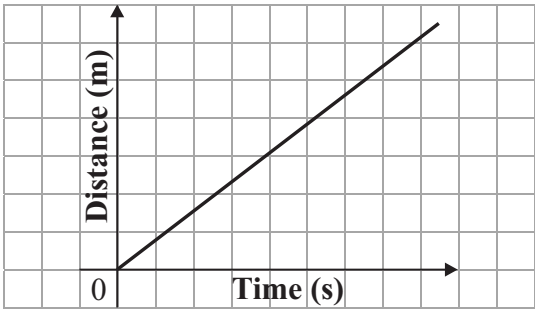
Question 6

(Suggested maximum time: 5 minutes)

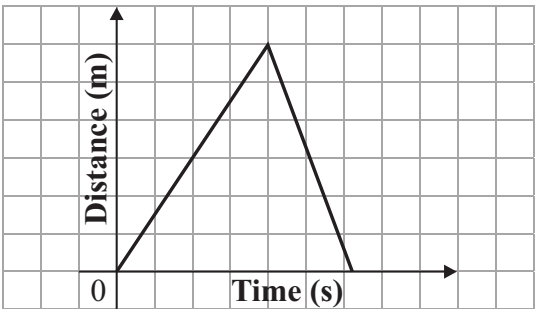
The speed of a number of different objects is monitored over a period of time.
The results are shown in the graphs below.



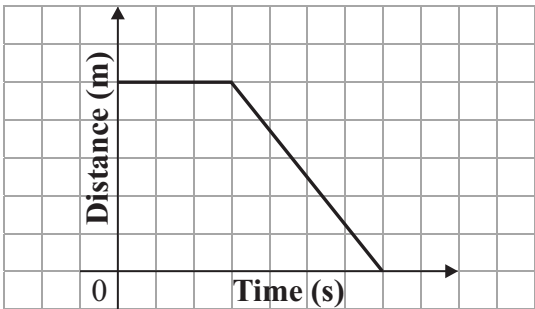
Graph A



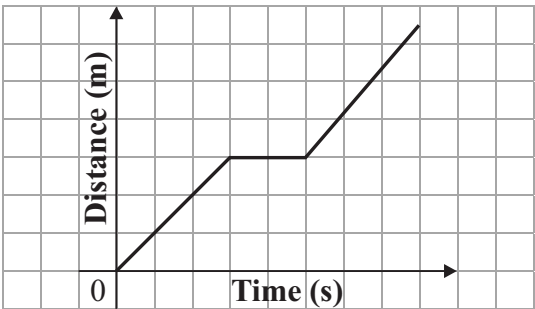
Graph B



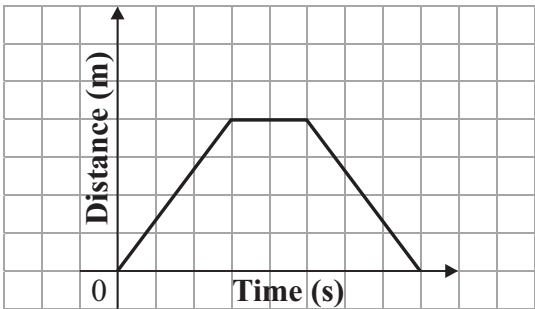
Graph C



Graph D



Graph E



Graph F

Write the letters A, B, C, D, E and F into the table to match the description with the graphs above.

Description	Graph
The speed starts at 0 m/s and increases continuously.	
The speed is constant for a time, then decreases.	
The speed increases quickly, then decreases quickly.	
The speed is constant.	
The speed increases, stays constant for a while, then decreases.	
The speed increases, stays constant for a while, then increases again.	

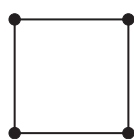
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Question 7

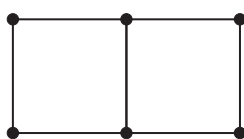
(Suggested maximum time: 10 minutes)

The first three stages of a pattern are shown below.

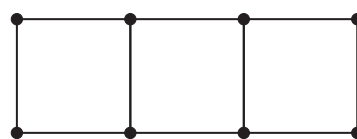
Each stage is made up of a certain number of dots and a certain number of lines which form squares.



1st stage

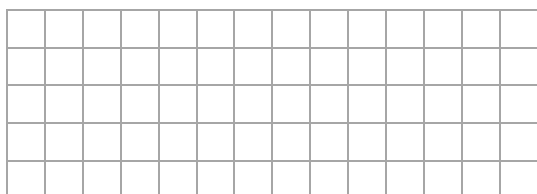


2nd stage

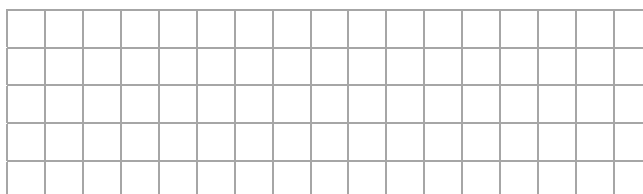


3rd stage

- (a) Draw the next two stages of the pattern in the grids below.



4th stage

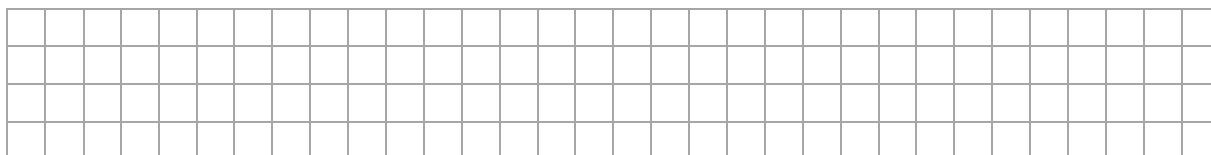


5th stage

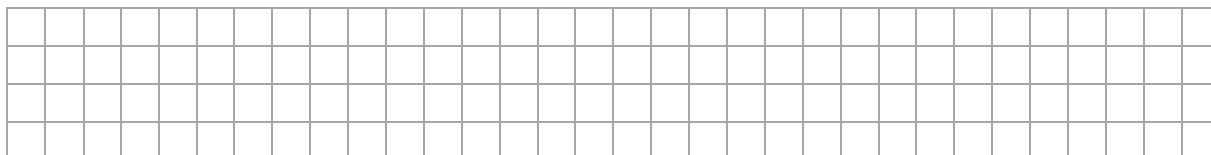
- (b) Complete the table below showing the number of dots and the number of lines required for the first six stages of the pattern.

Stage Number	1	2	3	4	5	6
Number of dots	4	6	8			
Number of lines	4	7	10			

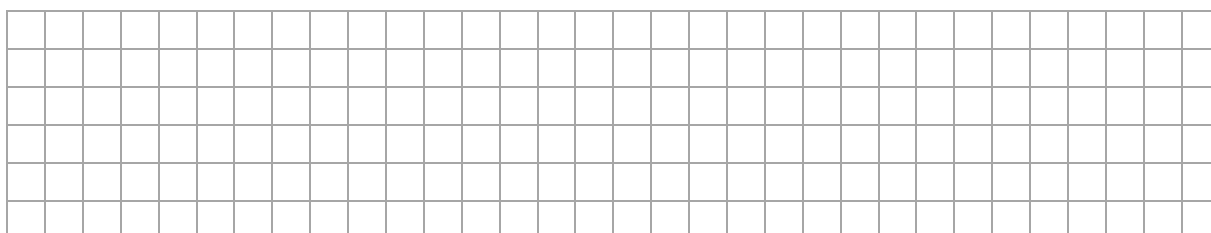
- (c) State the 'term to term' rule for the number of dots and the number of lines required for each stage.



- (d) How many squares can be made with 40 lines?



- (e) In a particular stage of the pattern, there are 22 dots.
How many lines are there in this stage of the pattern?



Question 8

(Suggested maximum time: 15 minutes)

- (a)** Factorise fully each of the following.

(i) $6xy + 3y$

[illegible]

(ii) $ab - bt + sa - st$

[illegible]

(iii) $x^2 - 36$

[illegible]

(iv) $x^2 - 7x + 12$

[illegible]

- (b)** Multiply $x + 2$ by $3x - 5$.
Write your answer in its simplest form.

[illegible]

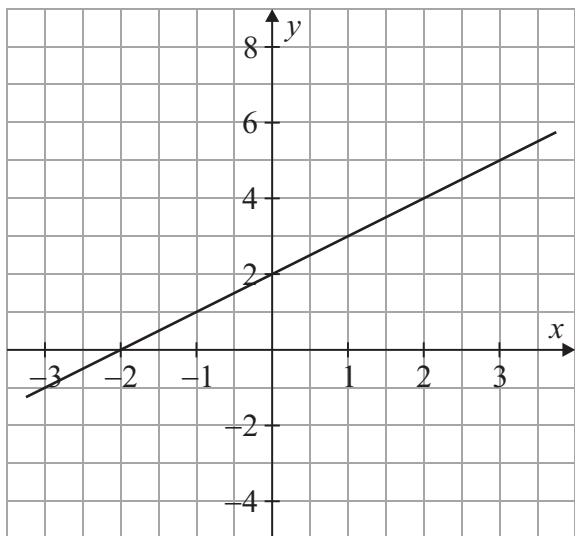
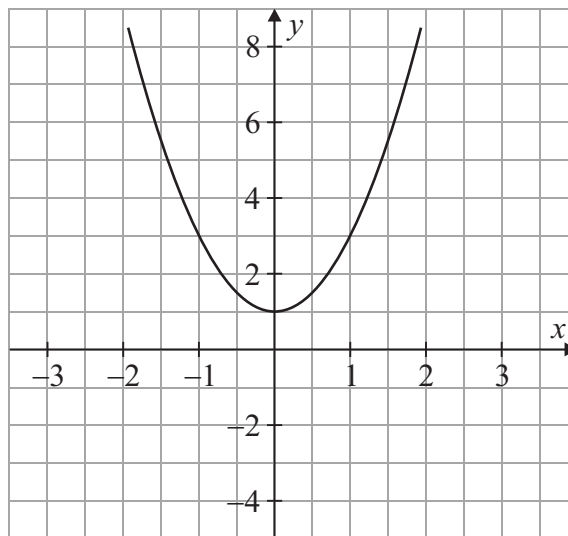
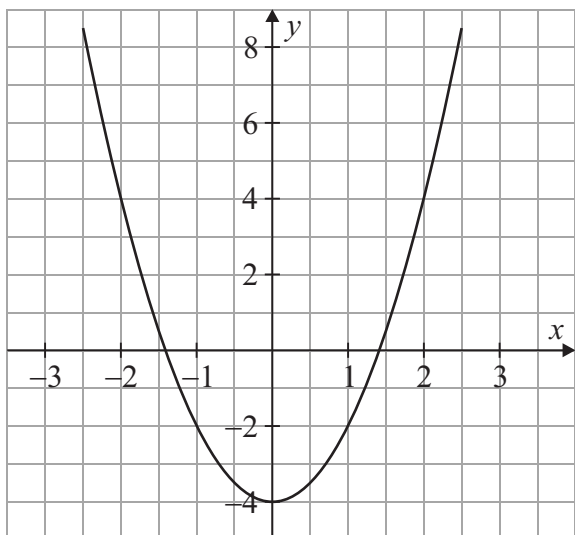
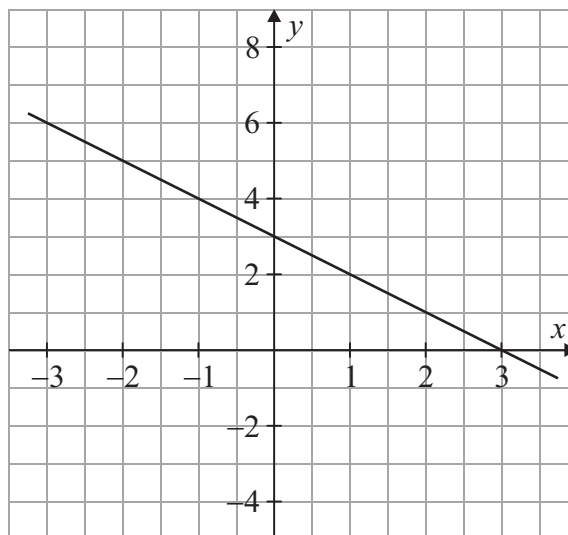
- (c) Divide $x^2 + 8x + 15$ by $x + 3$.

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Question 9**(Suggested maximum time: 5 minutes)**

The graphs of four functions are shown below.

**Graph A****Graph B****Graph C****Graph D**

Write the letters **A**, **B**, **C**, and **D** into the table to match each function with the correct graph.

Function	Graph
$y = x^2 + 1$	
$y = x + 2$	
$y = -x + 3$	
$y = x^2 - 4$	

Question 10

(Suggested maximum time: 15 minutes)

- (a) Put a tick (✓) in the correct box to indicate whether or not each of the following set of couples represents a function. Give a reason for your answer.

(i) $\{(6, 1), (4, 2), (6, -3), (2, 5)\}$

Function

5

Not a Function

Reason:

- (ii) $\{(5, 8), (3, -2), (-2, -5), (0, 0)\}$

Function

5

Not a Function

Reason:

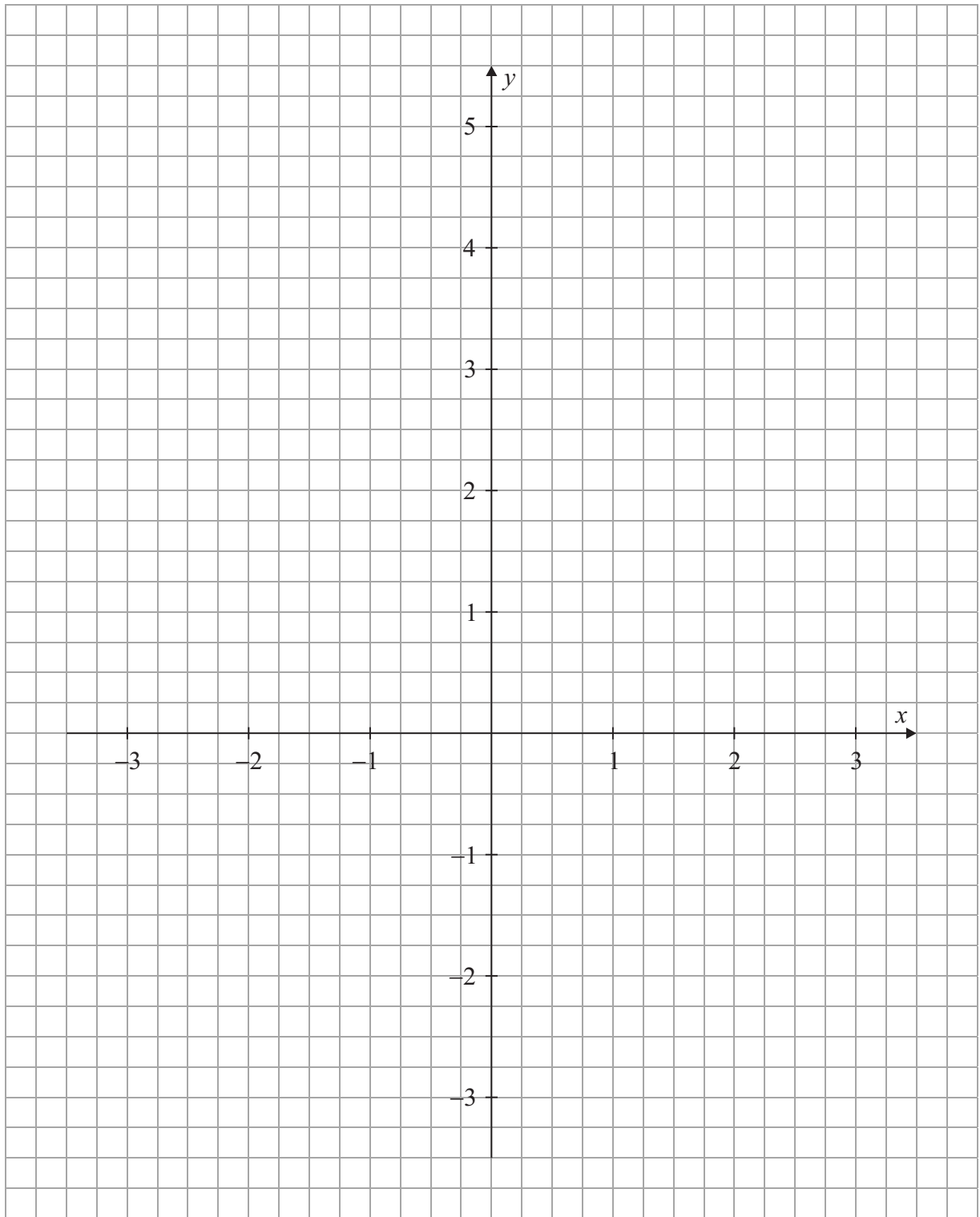
- (b) (i)** Complete the following tables for the functions $f: x \mapsto x + 1$ and $g: x \mapsto x^2 - 1$ in the domain $-2 \leq x \leq 2$, where $x \in \mathbb{R}$.

x	$f(x)$
-2	-1
-1	
0	
1	
2	

x	$g(x)$
-2	3
-1	
0	
1	
2	

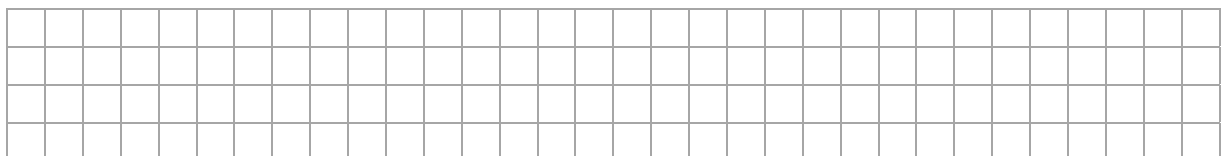
[illegible]

- (ii) Using the values in the tables, draw the graphs of the functions $f: x \mapsto x + 1$ and $g: x \mapsto x^2 - 1$ in the domain $-2 \leq x \leq 2$, where $x \in \mathbb{R}$.



- (iii) Use your graph above to write down the co-ordinates of the points where $f(x) = g(x)$.

Answer = (,) and (,)



Question 11

(Suggested maximum time: 10 minutes)

- (a)** Solve the equation $4(x - 2) = 12$.

- (b)** Write an algebraic expression for each of the following sentences. Use the letter x to represent the missing number. The first one is done for you.

- (i) Double the number and add 5.

$2x + 5$

- (ii) Triple the number and subtract 7.

- (iii)** Multiply the number by 8 and add 9.

- (c) In the square below, each **row** adds up to **24**.
For example, $a + a + a = 24$.

a	a	a
$2b$	a	a
a	b	$2c$

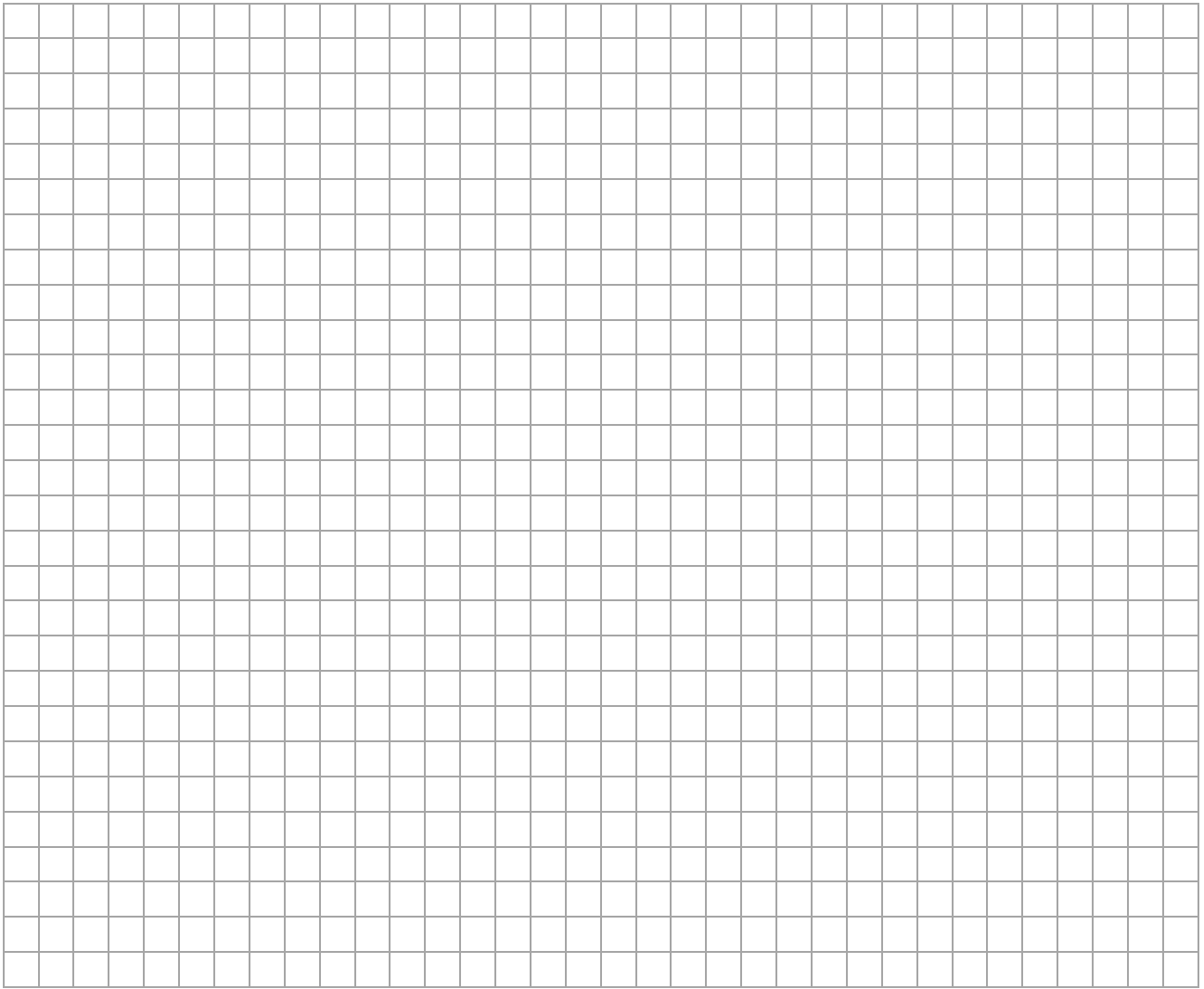
Find the value of a , the value of b and the value of c .

$a =$

$$b =$$

$$- \mathcal{C} =$$

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Pre-Junior Certificate, 2016 – Ordinary Level

Mathematics – Paper 1

Time: 2 hours

